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Synthesis of shape and structure-dependent hydroxyapatite nanostructures as a superior adsorbent for removal of U(VI)

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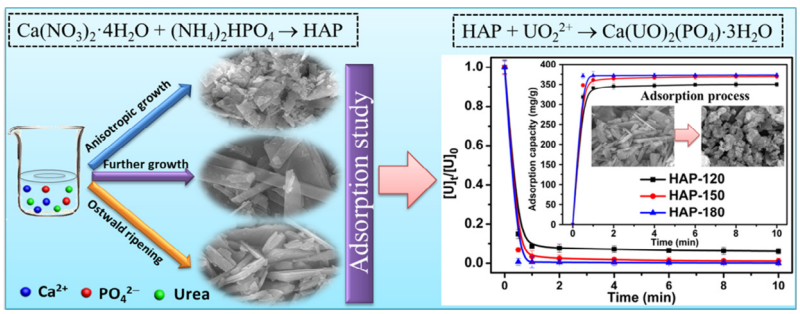
HIGHLIGHTS

- Shape and structure-dependent HAP nanostructures were hydrothermally synthesized.
- The temperature was a vital factor affecting the morphology and structure of HAP.
- Growth mechanism involved anisotropic growth, further growth and Ostwald ripening.
- HAP-180 exhibits excellent performance on removal of radioactive uranium.
- Removal mechanism involved adsorption, complex reaction and precipitation.

GRAPHICAL ABSTRACT



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ABSTRACT

Highly efficient removal of U(VI) from uranium-containing wastewaters is urgently needed with the global nuclear energy exploitation. Herein, we design a large-scale fabrication of shape and structure-dependent hydroxyapatite (HAP) nanostructures produced using a facile hydrothermal method in absence of any organic solvents and templates. It was found that the hydrothermal reaction temperature was a vital factor affecting the morphology (turned from nanosheets to nanoribbons and finally to nanoblocks) and structure (turned from monetite to hydroxyapatite and from calcium hydrogen phosphate hydrate to hydroxyapatite) of HAP. Moreover, the obtained HAP-180 exhibits highly efficient removal of U(VI) (99.3% removal of U(VI) within 30 s), and with the maximum adsorption capacity of 2024 mg/g, which is higher than those of HAP-120, HAP-150 and other materials previously reported. The superior removal performance is ascribed to the specific structure of HAP-180 and effective complexation with U(VI) ions. The findings of this work indicate that the morphology and structure of HAP can be feasible adjustment, and it is promising as environmental remediation material for U(VI) clean-up.

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1. Introduction

Uranium, a major byproduct produced from nuclear industry, has been defined as a toxic and radioactive heavy metal by the US Environmental Protection Agency (USEPA) [1–3]. The maximum uranium concentration level in drinking water set by the USEPA is 20 μg/L [4,5]. In recent years, with the vigorous development of nuclear power plants, a large amount of uranium-containing wastewater has been produced [6,7]. Uranium (half-life: 4.468×10^9 years) is one of the long-lived radionuclides [8]. Once the uranium-containing wastewater was discharged into aquatic environment without proper treatment, the released uranium may eventually store in the food chain and be ingested by humans, causing severe liver and kidney damage, and eventually death [9,10]. Hence, the removal of U(VI) from effluent have become a worldwide environmental concern. Adsorption is one of the attractive methods for the removal of metal ions from contaminated aqueous medium in terms of simplicity, suitability for ultra-low concentration, and availability of different low-cost adsorbents [11,12].

Many low-cost adsorbents have been developed for the removal of U(VI), including chitosan [1], modified silica [13,14], biomass [15], waste fish bone [16], metal organic framework [17], hydroxyapatite (HAP) [18–20], and so on. Of them, HAP is one of the most attractive material for U(VI) removal owing to the physical, chemical, mechanical and biological properties that make it stable and exhibiting high removal capability, and the presence of phosphate in HAP structure which constrain the mobility and transformation of U(VI) [21,22]. The longevity of U-P containing minerals precipitated has been verified by many researches [23,24]. The application of HAP has been investigated intensively, especially as reinforcement in collagen scaffolds for bone regeneration and removal of radioactive environmental toxic substances. For example, Ying et al. reported that the synthesis of ultralong HAP micro/nanoribbons can be used as reinforcement in collagen scaffolds for bone regeneration, and the ultralong HAP reinforced collagen scaffold possesses a porous structure, good flexibility and elasticity [25]. Wu et al. found that the fabrication of hierarchical hollow hydroxyapatite microspheres can be developed as superior adsorbents for removal of U(VI) ions, and the maximum adsorption capacity reached 199 mg/g [26]. However, the growth mechanism of HAP and its removal efficiency of U(VI) at different phases are seldom reported. Hence, it is extremely desirable to better understand the growth mechanism and design the fabrication of HAP.

Herein, we present a green technique without any organic solvents and templates for the synthesis of HAP samples with different morphology and structure. HAP sample prepared at 120 °C was mainly composed of $CaHPO_4$, $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ and $Ca_{10}(PO_4)_6(OH)_2$, and the morphology of HAP-120 was irregular and rough nanosheets. By elevating the temperature to 150 °C, the ratio of $CaHPO_4$ and $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ decreased considerably, while $Ca_{10}(PO_4)_6(OH)_2$ phase increased tremendously, and long HAP-150 nanoribbons with a length of about 10 μm and width of about 1 μm were obtained. When the hydrothermal temperature reached 180 °C, the HAP sample completely translated into $Ca_{10}(PO_4)_6(OH)_2$ and the morphology of HAP-180 was blocky with a length of about 2.2 μm, width of about 1.6 μm and thickness of about 150 nm. After that, the performance of HAP samples were evaluated through interacting with U(VI) in simulated wastewater stream. Impressively, HAP-180 exhibits highly efficient removal of U(VI) with the maximum adsorption capacity of 2024 mg/g and the faster removal rate of removal 99.3% within a short time interaction (i.e., 30 s). These findings may provide new insights into the design and development of versatile HAP materials with superior performance for the remediation of uranium contaminated medium.

2. Experimental

2.1. Reagents

Calcium nitrate tetrahydrate ($Ca(NO_3)_2{\cdot}4H_2O$, ≥ 99.0%), diammonium hydrogen phosphate ($(NH_4)_2HPO_4$, ≥99.9%), urea (CN_2H_4O, ≥99.0%), nitric acid (HNO_3, 65–68%), and ammonium hydroxide (NH_4OH, 25–28%) were purchased from Tianjin Fuchen Chemical Reagent Factory. Uranyl nitrate hexahydrate ($UO_2(NO_3)_2{\cdot}6H_2O$, ≥99.9%) was purchased from Hubei Chushengwei Chemical Co., Ltd. Various concentration of uranium stock solution (mg/L) were prepared by dissolving ($UO_2(NO_3)_2{\cdot}6H_2O$) in HNO_3 solution (pH = 3.0). All chemical reagents used in this study were of analytical grade and utilized without further purification.

2.2. Synthesis of shape-dependent HAP nanostructures

Shape-dependent HAP nanostructures were synthesized via a facile hydrothermal method. First, 0.1 mol/L $Ca(NO_3)_2{\cdot}4H_2O$ (solution A, 100 mL), 0.06 mol/L $(NH_4)_2HPO_4$ (solution B, 100 mL) and 1.0 mol/L CN_2H_4O (solution C, 40 mL) were prepared separately. Next, solution B and solution C were added dropwise into solution A orderly and the solution pH of mixture was maintained at 4.0 by adding diluted HNO_3. After vigorously stirring for 30 min, the mixture was then transferred into a 100 mL Teflon-lined stainless steel autoclave and hydrothermal treatment at 120 °C for 3 h. Finally, the resulting sample was collected by vacuum filtration and washed with DI water and ethanol, and dried in an oven at 80 °C for 10 h. The sample was denoted as HAP-120. For comparison, HAP-150 and HAP-180 were also synthesized using the same procedures except for the reaction temperature were 150 °C and 180 °C, respectively.

2.3. Characterization

The crystalline phases of the samples were determined by X-ray powder diffraction (XRD, PANalytical, PW3040/60) with Cu Kα radiation (λ = 1.5418 Å, 2θ = 5–80°). The functional groups of the samples were obtained by Fourier transform infrared spectroscopy (FT-IR, Bruker, Tensor 27) with the range of 400–4000 cm^{-1}. The surface morphologies and inner structures of the samples were examined by Field emission scanning electron microscopy (FE-SEM, JEOLJSM-7001F), and Transmission electron microscopy (TEM, JEM2010-HR). The surface areas and pore size distributions of the samples were conducted by Micromeritics ASAP 2460 instrument (USA). The values of pH at point of isoelectric of the samples were characterized by Zeta potential analyzer (Zeta PALS, Brookhaven, America), and the surface element compositions and valence states of the samples were analyzed by X-ray photoelectron spectroscopy (XPS, ESCALab250).

2.4. Batch adsorption experiments

The details of the batch adsorption experiments were given in Supplementary Material.

3. Results and discussion

3.1. XRD and FTIR analysis

The XRD patterns of HAP-120, HAP-150 and HAP-180 samples are shown in Fig. 1a. The HAP-120 sample was mainly composed of $CaHPO_4$ phase (monetite: PDF#71-1759), $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ phase (calcium hydrogen phosphate hydrate: PDF#74–1301) and $Ca_{10}(PO_4)_6(OH)_2$ phase (hydroxyapatite: PDF#74-0565). As to HAP-150, the intensity of diffraction peaks of $CaHPO_4$ and $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ decreased considerably, while those of $Ca_{10}(PO_4)_6(OH)_2$ increased tremendously. For HAP-180, only

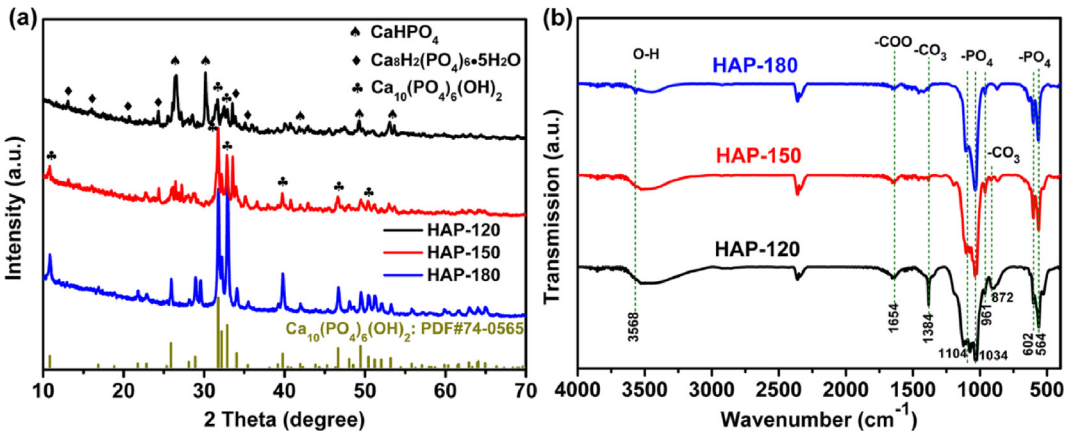

Fig. 1. (a) XRD patterns and (b) FT-IR spectra of HAP-120, HAP-150 and HAP-180 samples.

$Ca_{10}(PO_4)_6(OH)_2$ phase was observed. The characteristic peaks located at 2θ = 10.8°, 25.9°, 31.8°, 32.9°, 39.8°, 46.7° and 49.5° can be assigned to (1 0 0), (0 0 2), (2 1 1), (3 0 0), (1 3 0), (2 2 2) and (2 1 3) planes of the hexagonal structure of hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$: PDF#74–0565), respectively. As can be seen from the FT-IR results (Fig. 1b), the peak at 3568 cm^{-1} can be attributed to stretching vibration of the O–H group [27]. The peak at 1654 cm^{-1} is related to the COO^{-1} group, and the peaks at 1384 and 872 cm^{-1} are assigned to the CO_3^{2-} group [28–30]. The peaks at 1104, 1034 and 961 cm^{-1} are ascribed to the phosphate stretching vibrations, and the peaks at 602 and 564 cm^{-1} are attributed to the phosphate bending vibrations [31–33]. Noted that the HAP-120 exhibits the poor intensity of PO_4^{3-} groups, and high intensity of COO^{-1} and CO_3^{2-} groups, while the HAP-180 shows the high intensity of PO_4^{3-} and O–H groups, illustrating that the pure hydroxyapatite was successfully synthesized.

3.2. Morphology and microstructure analysis

The morphology and microstructure of the as-prepared samples were analyzed by SEM and TEM. Obviously, irregular and surface rough HAP-120 nanosheets were firstly formed at low hydrothermal temperature of 120 °C (Fig. S2a and b), implying that HAP-120 did not crystallize completely which is most likely due to the insufficient energy input. When the hydrothermal temperature was 150 °C, HAP-150 nanoribbons with a length of about 10 μm and width of about 1 μm were obtained (Fig. S2c and d). Further to increase of the hydrothermal temperature to 180 °C, blocky HAP-180 with a length of about 2.2 μm, width of about 1.6 μm and thickness of about 150 nm was successfully synthesized (Fig. S2e and f). These results suggest that the morphology of HAP altered remarkably when the hydrothermal temperature changed.

Fig. 2 exhibits TEM and HRTEM images of the obtained HAP samples that the morphology of HAP samples observed from the TEM images are consistent with the SEM results (Fig. 2a, c and e). From the HRTEM image of as-prepared HAP-120 sample, the lattice fringes with d-spacing of 0.348, 0.232 and 0.272 nm, corresponding to the spacing of (1 1 1) plane of triclinic structured $CaHPO_4$, (3 2 2) plane of triclinic structured $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ and (3 0 0) plane of hexagonal structured $Ca_{10}(PO_4)_6(OH)_2$, respectively (Fig. 2b). Fig. 2d shows the HRTEM image of HAP-150 sample, it can be observed that the inter-planar distance of fringes are 0.304, 0.280 and 0.351 nm, which are consistent with the (-2–11) plane of triclinic structured $CaHPO_4$, (−1 2 2) plane of triclinic structured $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ and (2 0 1) plane of hexagonal structured $Ca_{10}(PO_4)_6(OH)_2$, respectively. While the clear lattice fringes of HAP-180 with inter-planar distance of 0.281 and 0.343 nm corrospond to the spacing of hexagonal structured $Ca_{10}(PO_4)_6(OH)_2$, and there are no lattice fringes of other ingredient (Fig. 2f). These results suggest that the synthesis of HAP-120 and HAP-150 are composed of $CaHPO_4$, $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ and $Ca_{10}(PO_4)_6(OH)_2$, and the HAP-180 only composed of $Ca_{10}(PO_4)_6(OH)_2$, which is consistent with the XRD results.

The BET surface areas of HAP-120, HAP-150 and HAP-180 were calculated to be ca. 5.31, 9.42 and 10.79 $m^2\,g^{-1}$, respectively (Fig. 3a). This result suggests that the increase of hydrothermal temperature could improve the surface area of HAP. The pore size distributions curves obtained from the adsorption branch of the isotherms calculated by BJH model (Fig. 3b). As shown in Table S1, the average pores size of the HAP-120, HAP-150 and HAP-180 were 5.86, 5.99 and 5.99 nm, respectively.

3.3. Growth mechanisms of HAP samples

Based on the above results analysis, the possible growth mechanisms of as-prepared HAP samples are proposed and shown in Fig. 4. Firstly, small nuclei formed in the precursor solution, and then the nuclei gradually aggregated to form large nanosheets to reduce the total surface energy with the increase of temperature. The small nuclei and large nanosheets were mainly composed of $CaHPO_4$ and $Ca_8H_2(PO_4)_6{\cdot}5H_2O$ according to the XRD result. There were some small particles on the surface of nanosheets, according to thermodynamic theory, these small particles have high chemical potential owing to their little radius of curvature, thus, the next growth phase will preferentially implement on the small particles [34]. Meanwhile, with the consecutive decomposition of urea, the pH of precursor solution gradually increased and the following reaction occurred [35].

$$10Ca(NO_3)_2{\cdot}4H_2O + 8(NH_4)_2HPO_4 \rightarrow 2CaHPO_4 + Ca_8H_2(PO_4)_6{\cdot}5H_2O + 20NO_3^- + 16NH_4^+ + 4H^+ + 35H_2O \quad (1)$$

$$10CaHPO_4 + 12OH^- \rightarrow Ca_{10}(PO_4)_6(OH)_2 + 4PO_4^{3-} + 10H_2O \quad (2)$$

$$Ca_8H_2(PO_4)_6{\cdot}5H_2O + 3OH^- + 2Ca^{2+} \rightarrow Ca_{10}(PO_4)_6(OH)_2 + H^+ + 6H_2O \quad (3)$$

When the hydrothermal temperature reached 150 °C, decomposition of urea ensure the slow release of OH^- and improve the degree of supersaturation, which can make the HAP continue to grow in the longitudinal direction [25]. Thus, the small particles on the surface of nanosheets would disappear gradually, and nanoribbons with large size would form naturally. Further to increase of the hydrothermal temperature to 180 °C, the vertical and horizontal growth ratio of HAP will be reduced according to the Oswald ripening mechanism. Finally, the blocky HAP formed.

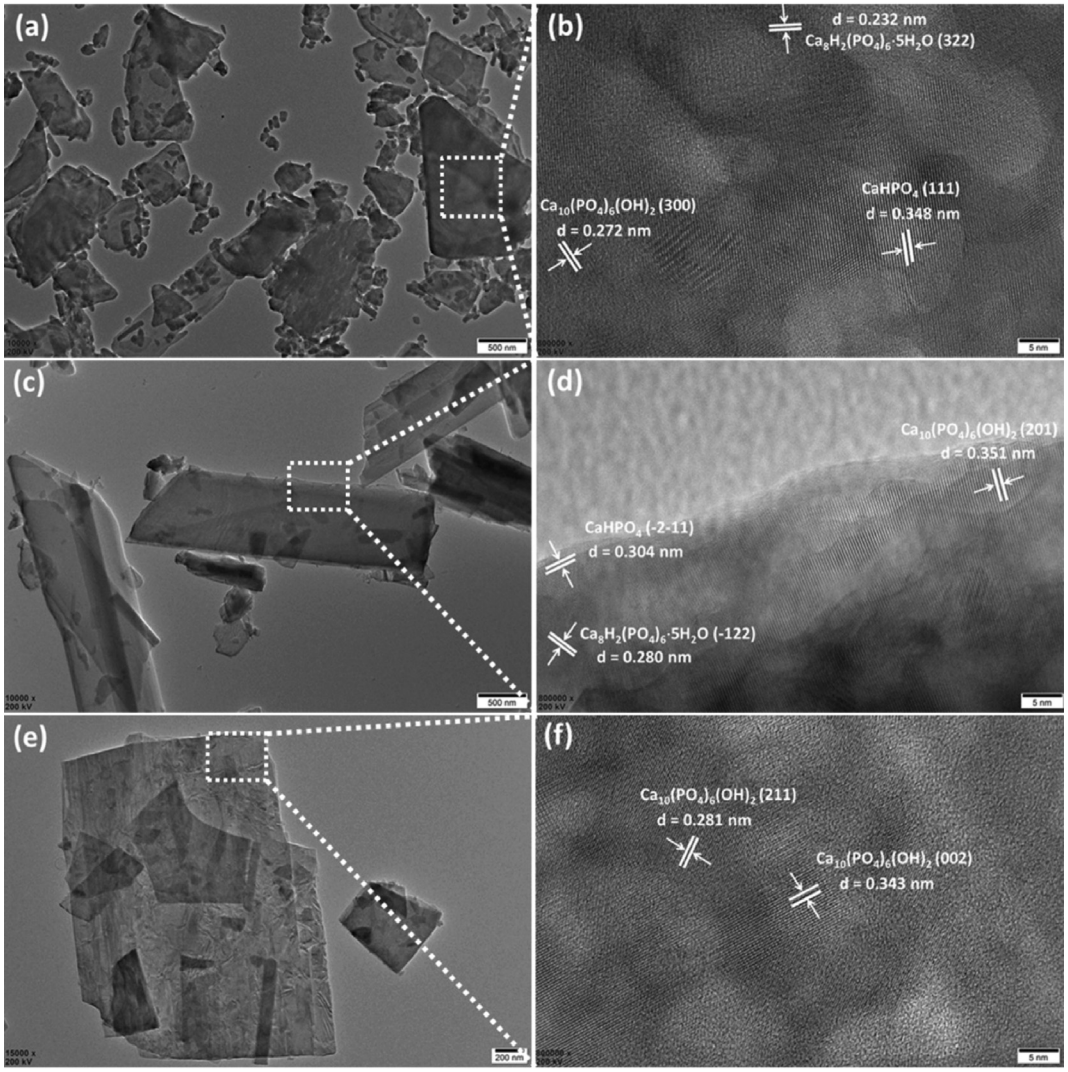

Fig. 2. (a) TEM image and (b) HRTEM image of HAP-120, (c) TEM image and (d) HRTEM image of HAP-150, (e) TEM image and (f) HRTEM image of HAP-180.

3.4. U(VI) removal by as-fabricated HAP samples

Fig. 5a shows that the removal efficiency of HAP-120, HAP-150 and HAP-180 toward U(VI) as function of time under constant stirring. Within 1 min, the U(VI) concentration in solution with HAP-180 has declined to 0.23 mg/L, After 5 min, the concentration of U(VI) in solution maintained at 0.10 mg/L. Under this circumstance, the U(VI) removal efficiency reached 99.7%. For comparison, the HAP-120 and HAP-150 show a low activity of U(VI) removal with the efficiency of 92.8% and 98.2%, respectively. The BET surface area-normalized adsorption capacities of as-prepared samples were calculated to be 65.88, 39.21 and 34.54 mg/m^2 for HAP-120, HAP-150 and HAP-180, respectively (Fig. 5b). These results suggest that the HAP-120 had the maximum removal density (mg/m^2) for U(VI). In order to investigate the optimal adsorbent dosage of HAP-180, the adsorbent dose experiments were conducted (Fig. 5c). It can be seen that the U(VI) removal

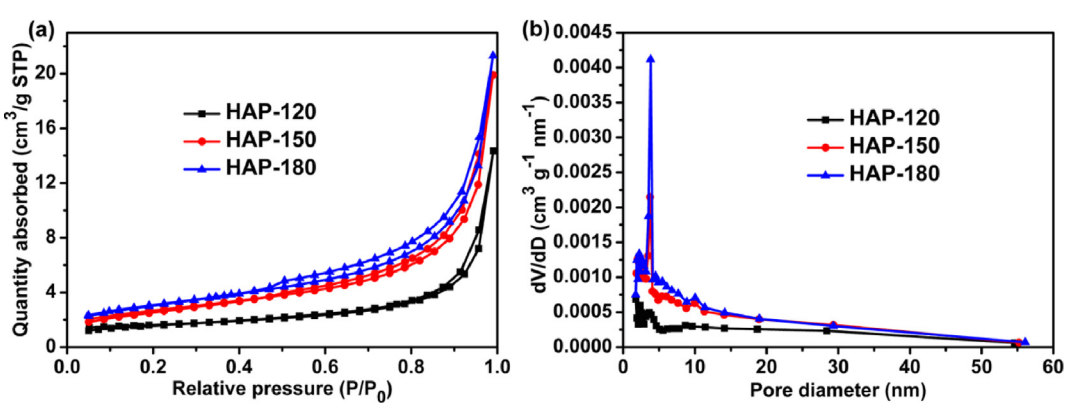

Fig. 3. Nitrogen adsorption–desorption isotherms (a) and pore size distributions (b) of the HAP-120, HAP-150 and HAP-180 samples.

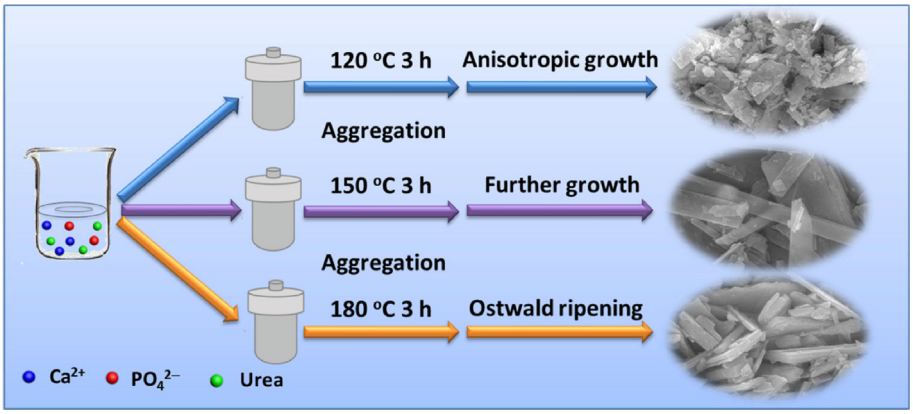

Fig. 4. Schematic presentation of the preparation of HAP nanostructures.

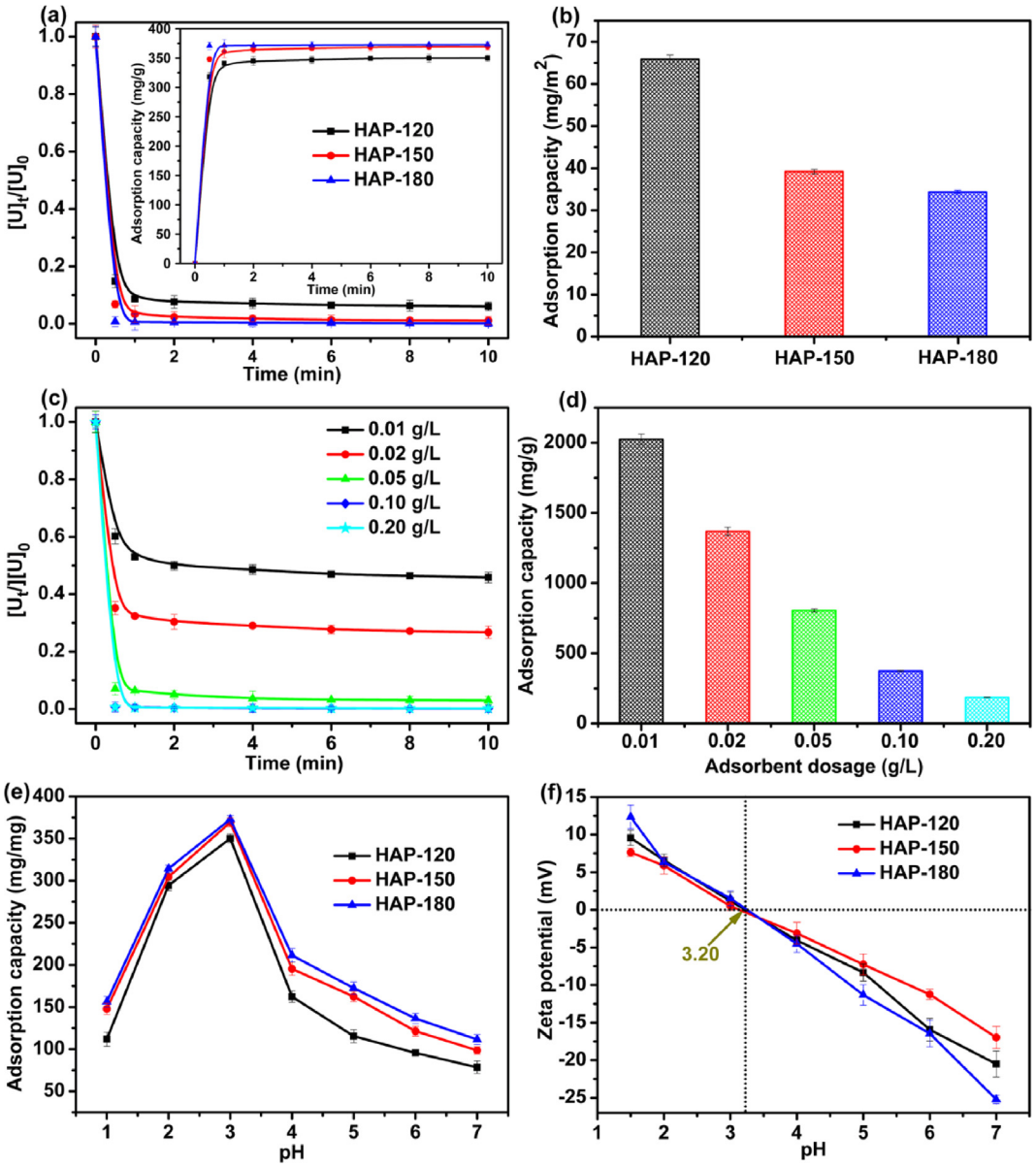

Fig. 5. (a) The effect of removal efficiency versus contact time (the inset is the effect of adsorption capacity versus contact time), and (b) The relationship of BET surface area-normalized adsorption capacity versus the types of HAP ($[U]_0$ = 37.36 mg/L, m/V = 0.10 g/L, pH = 3.0 and T = 298 K). (c) The effect of removal efficiency versus adsorbent dosage, and (d) The relationship of adsorption capacity versus adsorbent dosage (the adsorbent was HAP-180, $[U]_0$ = 37.36 mg/L, pH = 3.0 and T = 298 K). (e) The effect of adsorption capacity versus pH, and (f) The relationship of zeta potential versus pH ($[U]_0$ = 37.36 mg/L, m/V = 0.10 g/L and T = 298 K).

efficiency improved with the increase of HAP-180 sample. However, when the dosage of HAP-180 reached 0.10 g/L, the removal efficiency did not change significantly with the increase of HAP-180 dosage, which indicated that the optimal HAP-180 dosage was 0.10 g/L. Moreover, the adsorption capacity of HAP-180 was rapidly decreasing from 2024 mg/g to 186.4 mg/g with the increase of adsorbent dosage from 0.01 g/L to 0.20 g/L (Fig. 5d), suggesting that the as-prepared HAP-180 possesses an excellent adsorption capacity.

The solution pH is a key factor influencing the precipition of U(VI) ions and surface charge of adsorbents [36,37]. Therefore, the effect of solution pH on the sorption of U(VI) over as-prepared HAP samples were systematically investigated in the pH range of 1.0 to 7.0 (Fig. 5e). The pH of solution was adjusted by diluted HNO_3 and NH_4OH. It can be seen that the adsorption capacity of U(VI) increased with the pH improved from 1.0 to 3.0, while decreased with the pH increased from 3.0 to 7.0. Thus, at pH 3.0, the U(VI) adsorption capacity of as-prepared HAP samples reached the maximum. This is because that under low pH (pH < 3.0), the uranyl ions (UO_2^{2+}) are dominate in the solution of uranium [38,39]. Moreover, the isoelectric point (*iep*) of as-prepared samples were about 3.20 (Fig. 5f), thus producing electrostatic repulsion that hinder the surface complexation of U(VI) on the HAP surface, and result in low adsorption capacity. When the pH of solution was above 3.0, the charge of as-prepared HAP samples decreased and changed to negative with the pH increased higher than 3.20. But the uranyl ions start to hydrolyze and forming new substances, including $(UO_2)_2(OH)_2^{2+}$, $(UO_2)_3(OH)_5^{+}$, $(UO_2)_4(OH)_7^{+}$, $UO_2(OH)^{+}$ and $(UO_2)_3(OH)_7^{-}$, which are generally difficult to adsorb on HAP surface due to the larger U(VI) ionic radii and lower adsorption affinities than UO_2^{2+} [40].

3.5. Adsorption isotherms and kinetics

Adsorption isotherm studies were used to evaluate the nature of adsorption process [41,42]. Thus, two widely used adsorption isotherm models of Langmuir and Freundlich were chosen to analyze the adsorption process of U(VI) onto the as-prepared HAP samples. The nonlinear fitting curves of Langmuir and Freundlich isotherm models are displayed in Fig. 6a-b and the fitting parameters are shown in Table S2. It can be observed that the correlation coefficients (R^2) of HAP-120, HAP-150 and HAP-180 were 0.9787, 0.9949 and 0.9970 for Freundlich isotherm model; and 0.9239, 0.8093 and 0.8319 for Langmuir isotherm model, respectively. This indicates that the as-prepared HAP samples possess heterogeneous adsorption surfaces and the adsorption of U(VI) onto the HAP surface was multilayer adsorption process [43,44]. Fig. 6c and d exhibit the nonlinear fitting curves of pseudo-first-order and pseudo-second-order kinetic models and the fitting parameters are shown in Table S3. It can be seen that the correlation coefficients (R^2) of HAP-120, HAP-150 and HAP-180 were 0.9995, 0.9995 and 0.9998 for pseudo-first-order kinetic model; and 0.9995, 0.9999 and 0.9999 for pseudo-second-order kinetic model, respectively. Obviously, the values of the R^2 for pseudo-first-order kinetic model are consistent with pseudo-second-order kinetic model, indicating that the adsorption kinetics of U(VI) onto as-prepared HAP samples not only can be described by pseudo-first-order kinetic model, but also by pseudo-second-order kinetic model [45,46]. The calculated equilibrium adsorption capacity of U(VI) onto HAP-120, HAP-150 and HAP-180 were found to be 352.7, 370.1 and 372.7 mg/g, respectively. These values were in accordance with the adsorption experimental data above, and which is considerably higher than those of other adsorbents previously reported (Table S4).

3.6. Adsorption mechanisms

Fig. 7a and b show the surface morphology of HAP-180 before and after adsorption of U(VI). It can be seen that before the adsorption of U (VI), the as-prepared HAP-180 sample had a smooth surface and presented blocky structure with a length of about 2.2 μm, width of about 1.6 μm and thickness of about 150 nm. Interestingly, after adsorption of U(VI), the surface of HAP-180 became rough and irregular, and new block-like crystals were formed. Moreover, the color of the HAP-180 sample turned from pale white to yellow with the adsorption of U(VI) (Fig. S4). In order to confirm the ingredient of new block-like crystals, the X-ray diffraction pattern was employed to analyze the yellow sample (Fig. 7c). It can be observed that the new diffraction peaks located at 2θ = 10.5°, 16.4°, 17.9°, 20.9°, 24.6°, 25.5°, 27.6° and 34.3° were assigned to (0 0 1), (1 0 1), (1 1 0), (0 0 2), (1 0 2), (2 0 0), (2 0 1) and (1 0 3) planes of the tetragonal metaautunite ($Ca(UO)_2(PO_4)\cdot 3H_2O$: PDF#39–1351), respectively. The presence of metaautunite phase and absence of hydroxyapatite phase illustrated that the U(VI) was adsorbed and then reacted with hydroxyapatite to form the metaautunite. In addition, Fourier transform infrared spectroscopy was used to analyze the functional groups information of HAP-180 after adsorption of U(VI). As shown in Fig. 7d, a new peak located at 919 cm^{-1} appeared in the FTIR spectrum of U(VI) loaded onto HAP-180, representing the characteristic peak of UO_2^{2+} [47,48]. Moreover, the characteristic peaks intensity of PO_4^{3-} located at 564, 602, 961, 1034 and 1104 cm^{-1} were decreased significantly, which were mainly attributed to the PO_4^{3-} groups were chemical binding of U(VI) [49]. These results suggest that the as-prepared HAP-180 can immobilization of U(VI) with long-term stability.

X-ray photoelectron spectroscopy (XPS) was further used to prove the adsorption mechanism of U(VI) loaded onto HAP-180. A typical XPS survey spectrum of HAP-180 before and after adsorption of U(VI) are shown in Fig. S5, it can be observed that the presence of Ca, P and O elements in the HAP-180 sample, and three new strong double peaks ascribed to antisymmetric vibration of UO_2^{2+} were detected for HAP-180 adsorption of U(VI), which is consistent with the EDS elemental mapping results (Fig. S6). Two broad peaks of Ca 2p at 347.35 eV (Ca $2p_{3/2}$) and 350.92 eV (Ca $2p_{1/2}$) are attributed to the calcium compound (Ca^{2+}) (Fig. 8a) [31]. The peaks at 133.41 eV (before adsorption) and 133.49 eV (after adsorption) are ascribed to the PO_4^{3-} in $Ca_{10}(PO_4)_6(OH)_2$ and $Ca(UO)_2(PO_4)\cdot 3H_2O$, respectively (Fig. 8b) [50]. For the spectra of O 1 s (Fig. 8c), the peak at 530.8 eV can be attributed to the anion oxygen (O^{2-}), the peak at 531.4 eV can be ascribed to the hydroxyl groups (−OH), and the peak at 532.1 eV can be assigned to the adsorbed water on the surface [51,52]. It can be seen that the area ratio of − OH decreased from 44.22% to 39.02%, indicating the − OH plays a vital role in the adsorption of U(VI) [53]. Meanwhile, the fitting results of U $4f_{7/2}$ and U $4f_{5/2}$ indicated that the peaks of U(VI) appeared at approximately 382.2 eV and 393.1 eV, respectively (Fig. 8d), suggesting U(VI) adsorption on HAP-180 did not involve in reduction reaction [54].

4. Conclusions

In summary, shape and structure-dependent HAP nanostructures were successfully fabricated by a facile organic solvent-free and template-free hydrothermal method. The temperature of the hydrothermal system plays a vital factor for the successful synthesis of HAP nanostructures with tunable morphology and structure. The possible formation mechanism of the as-prepared HAP samples involves the processes of nucleation, aggregation, anisotropic growth, further growth and Ostwald ripening. In addition, the adsorption experimental results indicated that the HAP-180 had the maximum adsorption capacity of 2024 mg/g, which is higher than those of HAP-120, HAP-150 and other materials previously reported. The procedures of as-prepared HAP removal of U(VI) was that: firstly, the U(VI) was adsorbed on the surface of HAP, then HAP and U(VI) complex reaction, and finally forms the stable metaautunite material. Therefore, this study provides the pivotal information for understanding the fabrication of HAP samples with tunable morphology and structure, and also sheds light on the environmental remediation applications of HAP materials.

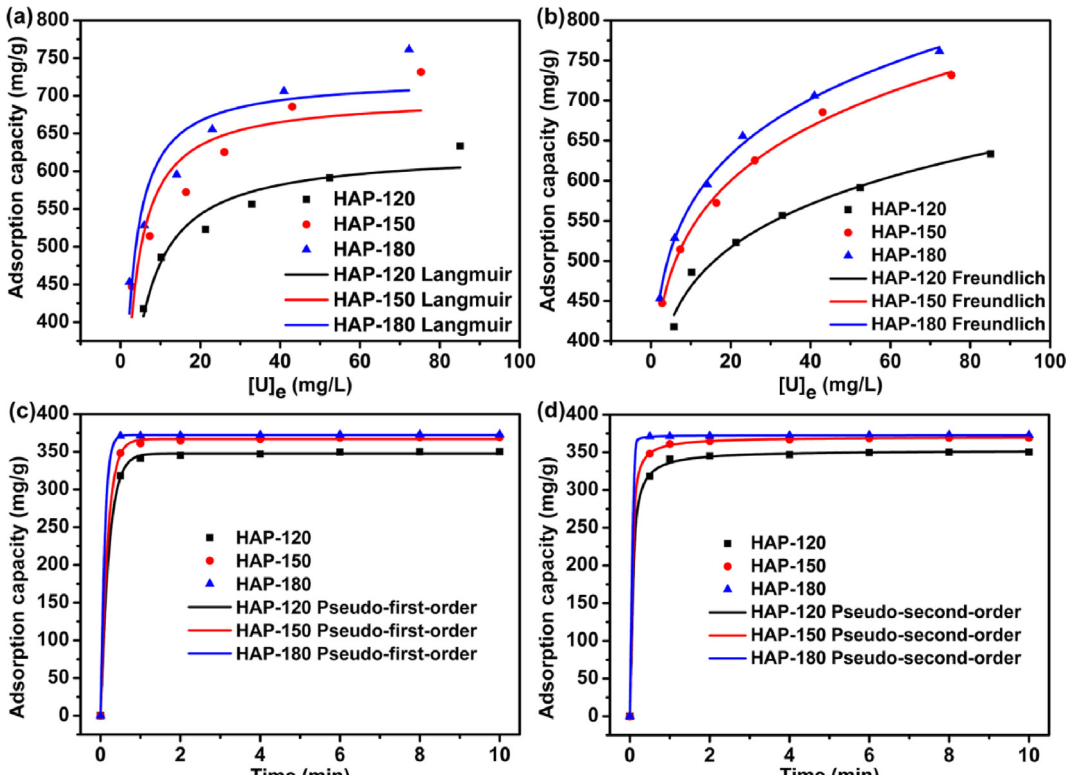

Fig. 6. (a) Nonlinear Langmuir isothermal model and (b) Nonlinear Freundich isotherm model of HAP-120, HAP-150 and HAP-180 samples. (c) Nonlinear pseudo-first-order kinetic and (d) Nonlinear pseudo-second-order kinetic of HAP-120, HAP-150 and HAP-180 samples.

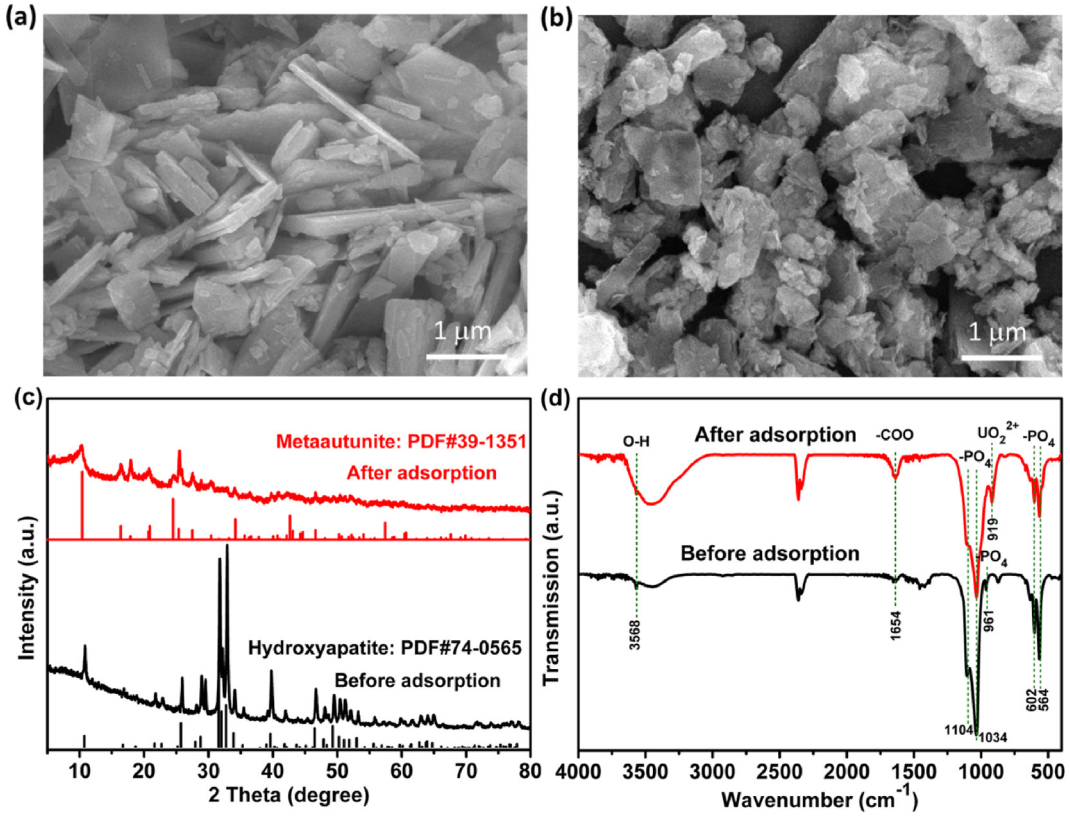

Fig. 7. SEM images of HAP-180 before (a) and after (b) adsorption of U(VI). XRD patterns (c) and FTIR spectra (d) of HAP-180 before and after adsorption of U(VI).

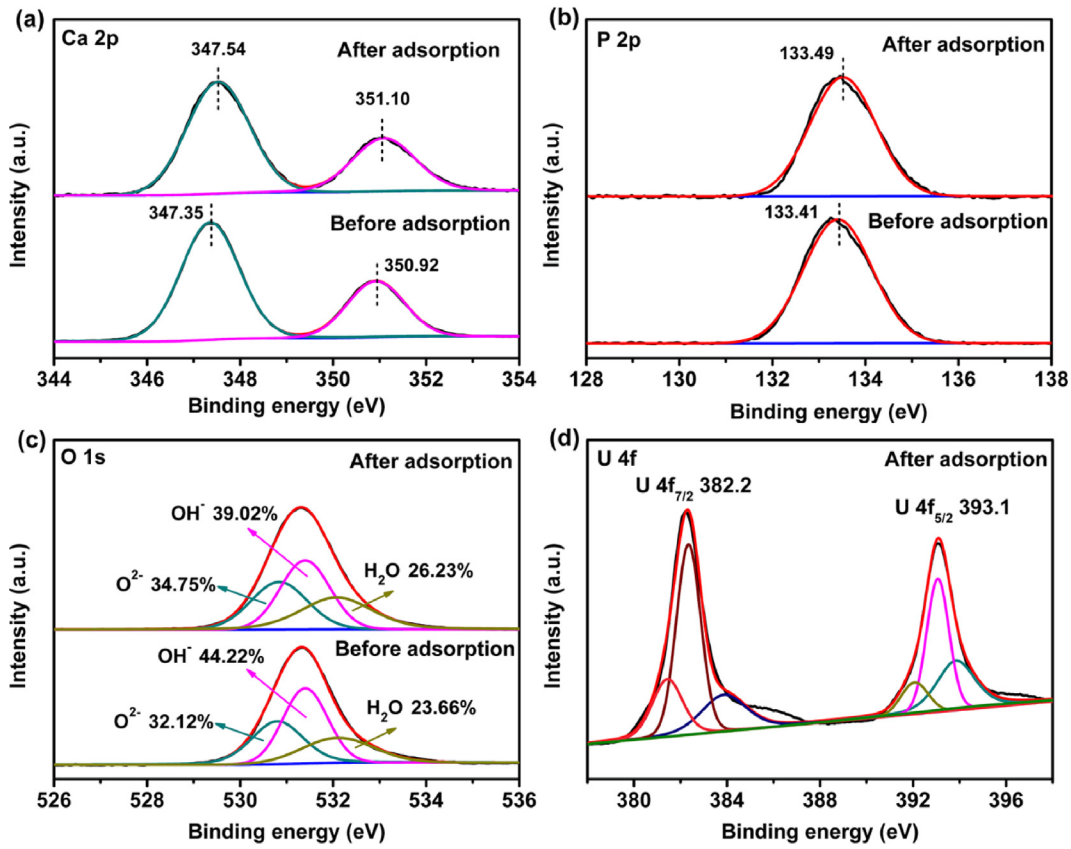

Fig. 8. High-resolution XPS spectrum of (a) Ca 2p, (b) P 2p, (c) O 1 s and (d) U 4f of before and after adsorption of U(VI) onto HAP-180.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2019.123262.

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